

Review

Development and Validation of a Theoretical Model for Addressing Problems in Agile Meetings: A Systematic Literature Review and a Qualitative Study

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Abstract: Agile meetings are an essential part of modern software development, but they often face challenges such as lack of focus, communication problems, technological difficulties, and the like. Previous research has often addressed individual issues of agile meetings but has not provided sufficiently integrated or practical models to comprehensively address these challenges. This research fills this gap, developing a theoretical model that not only identifies key issues but systematically links them to provide a deeper understanding of their impact on meeting effectiveness. In this paper, we present a theoretical model that depicts intra-team problems in agile meetings developed through a systematic literature review (SLR). After initial development, the model was validated and improved through the qualitative analysis of semi-structured interviews with IT experts. This research aimed to confirm the coverage and relevance of the content of the preliminary model, as well as to identify the need for modification. The results of the interviews suggest that the presented model is adequate but that there is a need for potential improvement. IT experts agreed with the items in the model, and some of them suggested improvements. The model is promising because it illustrates the key issues that influence the effectiveness of agile meetings in a promising way. IT experts agreed on the items in the model, and some of them suggested improving the model.

Keywords: agile meeting problems; theoretical model; systematic literature review; qualitative study



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1. Introduction

Agile methodologies emerged as a response to problems in traditional project management methods [1]. Originally introduced in the Manifesto for Agile Software Development in 2001, agile has set new standards for flexibility and adaptability in software development [2,3]. Agile methodologies are a set of iterative and incremental approaches to software development and project management. They prioritize flexibility, collaboration, and customer feedback to deliver high-quality products efficiently. Mention methodologies emphasize adaptive planning, continuous improvement, and rapid delivery of working software. Agile methods are very popular because of their features in software development [4].

Key principles of agile methodologies include [5]:

- Breaking the project into small iterations, allowing for frequent releases and feedback loops.
- Involving stakeholders throughout the development process to ensure alignment with their needs and priorities.
- Embracing change and adjusting plans based on feedback and evolving requirements.
- Bringing together individuals with diverse skills and expertise to promote collaboration and problem-solving.

- Reflecting on performance and processes to identify areas for enhancement and implementing changes incrementally.

Scrum is the most popular agile development methodology [5–8]. Scrum was developed to improve team collaboration and efficiency through structured processes and frequent evaluations [9]. Meetings in the Scrum methodology facilitate communication in the execution of tasks [10]. A daily fifteen-minute Scrum meeting is held by the manager and members of the development team, during which the manager asks questions to be discussed: What has been completed? What things need to be done, and what problems do the team members have? A sprint planning meeting is a monthly meeting attended by the product owner, Scrum master, and the team to discuss what will be done for the next sprint. The project is divided into small tasks to complete the tasks in one sprint. A sprint review meeting is a monthly meeting held at the end of a sprint. It is a time-limited meeting where team members present the incremental working model that has been completed. A sprint retrospective meeting is held after the completion of each sprint. Team members take appropriate action for the future sprint and determine if everything is fine and if some improvement is needed [10,11]. These are some types of meetings that are carried out, and there are more, depending on the company.

Scrum benefits [12]:

- Scrum supports teams to complete their products rapidly and coherently;
- Scrum guarantees efficient use of money and time;
- Big projects are divided into maintainable sprints;
- Developed products are coded and tested throughout the sprint analysis;
- Every task is done properly for rapid software development;
- The team members can achieve a clear idea with Scrum meetings;
- Get feedback from customers and stakeholders for better improvements;
- The self-effort of every team member is seen in every Scrum meeting.

Scrum teams are the core of the Scrum framework [7,13]. Meetings are conducted within the Scrum methodology. Meetings enable teams to coordinate, communicate, and adapt to rapidly changing project requirements and priorities. They are of great importance, and it is necessary to emphasize the problems in agile teams. Problems in agile teams include lack of focus, poor member engagement, technology issues, lack of transparency, conflicts, communication challenges, and information overload. Problems significantly affect team efficiency and productivity, hindering collaboration and quality of work [14,15].

In comparison to Scrum, the Kanban method is less detailed in its required practices and principles, making it a more flexible approach. While Scrum is known for its structured framework with specific roles and ceremonies, Kanban is less rigid and offers greater adaptability [16]. Despite this flexibility, Kanban is highly regarded for its ability to address challenges that other methodologies may not handle as effectively [17]. It is especially effective for teams aiming to deliver business value incrementally [17]. In practice, Kanban supports process evolution through several key practices. These include visualizing the workflow with a Kanban board, which enhances transparency, and limiting work in progress by reducing the number of features being implemented simultaneously. Kanban also focuses on managing and measuring flow to identify bottlenecks, making policies explicit to ensure clarity, and implementing feedback loops for ongoing refinement. Additionally, Kanban promotes collaborative and continuous improvement as integral aspects of the process [18]. Often, some agile methodologies are combined, depending on the needs of the company. Combining different agile methodologies is a common practice in the industry to adapt to the specific needs of projects and teams. These combinations may include the use of elements from multiple agile methodologies to achieve better efficiency and responsiveness to change. Scrumban is expected to override both Scrum and Kanban, as it inherits the best features of both. Scrumban incorporates the iterative planning of Scrum but is more responsive and adaptive to changes in user requirements [19]. There are other agile methodologies, but these three are most often applied in practice.

The paper initially presented a short systematic review of the literature (SLR). SLR allows us to identify problems in teams during meetings. The literature review provides insight into existing theories and models. This helps shape and develop the model. This review reveals the key issues and factors that influence agile methodologies and team meetings. Understanding these issues helps to refine the aspects to include in the model. After developing a theoretical model based on an SLR, qualitative research is used to gather additional insights and feedback that will enable further refinement and validation of the model. The systematic review provides the basis on which the qualitative research was built, where we asked relevant interview questions. SLR increases both the validity and reliability of research by drawing on a comprehensive review of existing research and theories. This helps strengthen the argumentation and foundation of the theoretical model.

After an SLR, a method of further research was chosen to enrich the model. The choice of method is an important aspect of the successful implementation of research. Qualitative research was chosen for this research because qualitative research helps with understanding the different perspectives and motivations of team members. This can reveal how and why certain problems occur and how team members experience them, which is crucial for adapting the model to their real needs [20,21]. Agile meetings and problems in teams often involve complex and dynamic phenomena that are difficult to quantify. Qualitative methods enable the investigation of these complex aspects through a detailed examination and analysis of the experiences and opinions of the participants [22,23]. Qualitative research enables a deeper understanding of problems that were not covered by previous research, and it is also useful for exploring and understanding complicated matters and refers to a research approach that involves data collection without quantitative measurement [24,25]. In this research, an approach for gathering information called an interview was used. Interviewing is an approach used to gather information from people [26]. It is a method of data collection that involves asking questions to elicit responses from research participants. Also, interviews are often used in software engineering research [27]. Interviews provide recommendations and solutions from IT experts who face similar problems in practice, and this contributes to the development of practical and applicable aspects of the model.

The questioning process can be such that the researcher formulates questions as they come to mind, or the researcher strictly adheres to pre-determined questions [28]. Kumar [26] states that when using a structured interview, the uniformity of responses is ensured. As the respondent answers more consistently, this could lead to irrelevance to the topic being measured. It may also show a loss of spontaneity [28]. In a structured interview, there is no room for variation in answers [29]. There is very little flexibility in how questions are asked or answered within a structured interview [30]. Interviewers should not engage in long studies so as not to deviate from the introduction to the study, and they should not suggest answers to respondents and similar [30]. The semi-structured interview is a popular method of data collection because it has proven to be versatile. It can be combined with individual and group interview methods [31], and the rigidity of its structure can vary depending on the purpose of the study and the research questions [32]. The semi-structured interview method is successful in enabling reciprocity between the interviewer and participant [33], enabling the interviewer to improvise follow-up questions based on the participant's responses [34,35]. Therefore, semi-structured interviews will be used as a data collection method in this research. The questions are prepared in advance, but there is room for improvisation and additional questions that arise during the conversation. Seven IT experts from the IT sector were interviewed, which enabled the collection of a rich and diverse set of data. The interviews were conducted in person or via the Zoom platform and lasted about an hour. This enabled the collection of detailed and qualitative insights into the problems faced by teams during agile meetings. Feedback from IT experts can reveal potential weaknesses or deficiencies in the model and help improve it. Also, qualitative insights can help formulate hypotheses and questions for future research, thereby contributing to a broader understanding and further exploration of the themes.

This paper aims to validate the coverage and relevance of the content of the preliminary model, as well as to identify the possible need to modify the presented model in which the agile meeting's problems in teams are illustrated. We offer a succinct summary of the systematic literature review (SLR), followed by an identification of the issues highlighted within the model. The presented model with observed problems was created by the authors for IT experts to point out additional problems, all to improve and enrich the model. Based on the literature analysis, it was seen that no work has provided a sufficiently unified solution that addresses the problems in agile meetings. Based on this, the goal is to develop a model that will identify key challenges in agile meetings and validate the model through expert feedback and future practical applications.

2. Related Work

Cockburn [36] emphasizes the importance of collaboration between team members as a key success factor in agile software development; emphasizes the importance of cooperation between team members who have different roles within the team; and points to the importance of strengthening the team through communication, trust, and openness.

Lencioni [37] emphasizes the importance of effective meetings. The paper describes research and solving problems that arise in connection with meetings in organizations. He also emphasizes that the identified problems can negatively affect the productivity and engagement of the team.

Hackman and Wageman [38] explore problems that can arise in team meetings. They mention that meetings are sometimes a source of problems if they are not well managed. The authors show how poorly conducted meetings can result in wasted time, decreased productivity, frustration, etc.

Paasivaara et al. [39] show the challenges and effectiveness of coordination between teams in large, globally distributed Scrum projects. The authors explore specific challenges arising from distributed teams in the context of the Scrum methodology. These include time zone differences, language barriers, cultural differences, and technical challenges in communication and collaboration.

In a more recent study, Mortada et al. [40] present the reasons why software teams deviate from the Scrum methodology. Implementing agile methodologies can be challenging and subject to modification. Reasons for deviations can be a lack of understanding of the Scrum approach, lack of support or training for teams, and specific project requirements that are not compatible with the Scrum methodology, as well as internal resistance to change.

Perkusich et al. [41] show the development of a model for problem detection in software development projects using the Scrum methodology. Implementing an agile methodology can be complex, and there can be various challenges. The authors develop a model that includes criteria or indicators that signal possible difficulties in implementing Scrum. Teams may have different challenges that may arise when applying the mentioned methodology, such as problems with time management, lack of transparency or communication, excessive complexity of the process, or lack of clearly defined roles and responsibilities.

Reginaldo et al. [42] explore the challenges organizations face when transitioning to using agile methodologies. The authors highlight challenges such as resistance to change, lack of understanding of agile principles and practices, lack of support or training for employees, technological challenges, etc. This is qualitative research; the authors used qualitative methods to collect data and improve the understanding of the experiences and perspectives of organizations undergoing agile transformation.

Kurapati et al. [43] explore the reasons why organizations decide to adopt or reject certain agile practices. This may include factors such as the perceived value of the practice, organizational culture, technological or operational barriers, experiences with previous projects, etc.

Almeyda's systematic mapping study [44] conducts a systematic literature mapping to investigate process improvement in software requirements engineering. Existing research, works, and practices in the area of process improvement related to requirements engineering in software development are reviewed.

Stray et al. [45] explore the importance of daily team meetings. This paper focuses on the analysis of how daily meetings function in practice and what their purpose is, their effectiveness, their impact on teamwork, and the like.

Thompson and Nguyen [46] present strategies and practices for improving the effectiveness of agile meetings based on the experiences of practitioners. They analyze how agile teams can optimize meetings, all to be more productive, efficient team members and more focused on achieving goals in software development.

Dingsøyr et al. [47] present a systematic review of the literature on agile software development, which can provide a solid foundation for modeling and implementing solutions in agile teams. It explores key aspects of agile development, which are relevant to understanding the challenges and solutions in that context.

A recent study introduces the Agile Team Effectiveness Model (ATEM), which addresses the unique dynamics of teamwork in agile software development. This model identifies three key coordinating mechanisms—shared mental models, communication, and mutual trust—and five critical components: shared leadership, team orientation, redundancy, adaptability, and peer feedback. These elements are posited to significantly influence team effectiveness in agile environments, thereby providing a framework for enhancing collaboration and addressing challenges in agile meetings [48].

These studies indicate the importance of cooperation, communication, and trust within agile teams, as well as the importance of effective meeting management to avoid problems such as loss of productivity and frustration. Challenges faced by distributed teams, such as language and cultural barriers, as well as technical problems in communication, were particularly emphasized. It is also pointed out that the transition to agile methodologies can be complex due to a lack of support, training, and resistance to change within organizations. Finally, models are developed for detecting problems and improving team effectiveness in agile environments, emphasizing coordination, shared mental models, communication, and trust as key factors for success.

3. Development of Theoretical Model Based on Systematic Literature Review

In this study, we conducted a concise, systematic literature review following the methodology proposed by [49,50], which led to the development of a theoretical model. The purpose of an SLR is to identify, evaluate, and interpret relevant research papers about a specific objective [50].

Earlier studies highlight the significance of challenges faced by employees during agile meetings within teams, underscoring the necessity of systematic literature reviews. In an earlier paper [51], an inductive thematic analysis of the data obtained through anSLR was performed to determine the topics that represent problems at agile meetings. In the mentioned work, it was possible to see how an SLR is carried out and how topics are determined based on qualitative data analysis, from which the authors obtained information about important problems at the meetings. The aim of the paper [52] was to present a conceptual model of problems in agile meetings and their relationship with organizational issues in the IT industry. Meeting problems were also identified through literature analysis, and the model was based on the interpretation of the identified problems and their relationships. Additionally, [53] presents a conceptual model highlighting problems in agile meetings and their association with project issues in the IT industry, where problems are identified through literature analysis.

Creating a visual representation of an SLR process involves breaking it down into its key components and stages. This visual representation in Figure 1 helps convey the methodology and workflow involved in conducting anSLR on meeting-related problems.

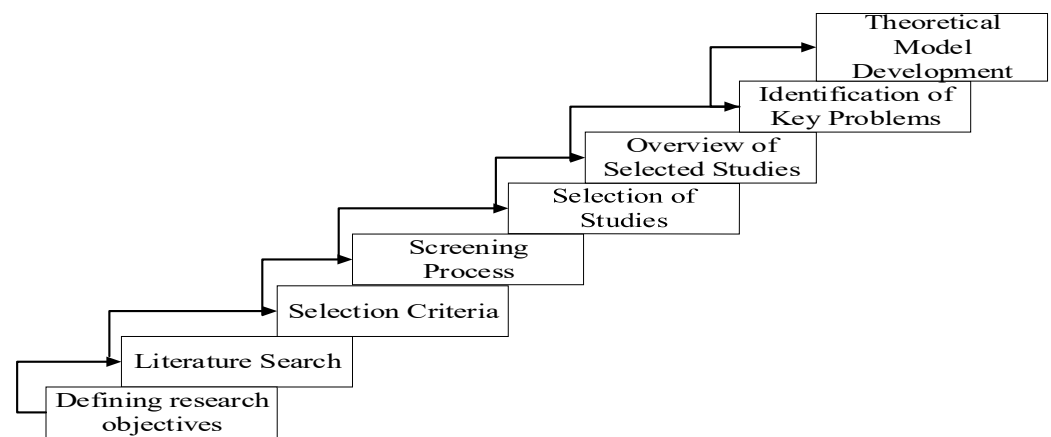


Figure 1. Developing a model using an SLR: process overview.

To meet the research objective, it was necessary to identify papers that documented team issues in agile meetings. To identify appropriate papers, we employed a literature search strategy centered around specific keywords to guide our search process. Search strategies [54] use alternative words and synonyms in each search string. The search string is based on keywords and logical connectors and addresses the following search string:

$$\{\text{agile meetings}\} \text{ AND } \{\text{team collaboration OR team communication or team dynamics}\} \text{ AND } \{\text{problems or obstacles OR challenges}\} \quad (1)$$

This combination was designed to capture a broad range of studies related to team issues in agile meetings, ensuring comprehensive coverage of the relevant literature. The following libraries were used for the literature review: IEEEEXPlore, Science Direct, Springer, and Wiley. After that, a review of the literature was also carried out in Google Scholar to increase the number of scientific papers. The aim was to review the most recent publications from 2011 to 2023, with a focus on conferences and journals in which the peer-reviewed papers were published. Papers were first selected by keyword criteria. Then, in the second selection process, papers were selected after reading the title, and the authors considered whether it suggested to them that the issues would be presented in the study. After that, in the next process of paper selection, the abstract was analyzed according to the same objectives, where a certain number of papers were selected. The authors read the introduction, body, and conclusion and considered whether the case studies were appropriately considering issues in agile meetings. To ensure that the primary studies were relevant to the aim of this review, the inclusion and exclusion criteria were as follows:

Inclusion criteria: (1) the works were published in the period from 2011 to 2023; (2) papers written in English; (3) papers that were published in reviewed journals and conference proceedings; (4) papers related to the IT and software industry; (5) papers that talk about agile meeting problems.

Exclusion criteria: (1) papers that are duplicates of papers that were already included; (2) papers not related to agile meeting problems; (3) unavailable PDF files.

Papers were selected if they met the inclusion criteria. After the inclusion of papers according to the criteria, the other parts of the paper were reviewed again, starting with the introduction and the conclusion sections. If the authors felt that the papers were relevant to this research, they selected those papers. The papers were selected by the authors, taking care as to whether the text of the paper had indications of problems from agile meetings that are exclusively related to problems in the team. This process was applied to each electronic library. The following section provides an overview of the selected studies.

Overview of the Selected Studies

Table 1 summarizes the activities undertaken for the literature search. Initially, 136 papers were identified across various scientific research libraries based on the formulated keyword search strategy. A review of the titles reduced this number to 62 relevant studies. Following this, the abstracts of the selected papers were analyzed, narrowing the selection to 31 papers. After applying specific inclusion and exclusion criteria, the number of relevant papers was further reduced to 26. A comprehensive review of the introduction, body, and conclusion of each of the 26 papers was conducted, ultimately leading to the selection of 16 scientific papers that explicitly addressed team issues in agile meetings. To ensure accuracy, a manual review was performed to verify that no duplicate entries were included in the final selection. Numbered lists can be added as follows:

Table 1. Selected studies.

Database	Searching Based on Keywords	Reviewing Title	Reviewing Abstract	Inclusion/Exclusion	Detailed Reading
IEEE Xplore	38	15	8	7	5
Science Direct	20	10	5	4	1
Springer	33	15	7	5	2
Wiley	5	2	1	1	1
Google Scholar	40	20	10	9	7
Total	136	62	31	26	16

Table 2 presents the primary studies that passed the selection process presented in Table 1.

Table 2. Primary studies.

Primary Studies (PS)	References
PS1	Stray, V.G.; Moe, N.B.; Aurum, A. Investigating daily team meetings in agile software projects. In 38th Euromicro Conference on Software Engineering and Advanced Applications, 2012. [46]
PS2	Stray, V.G.; Lindsjörn, Y.; Sjøberg, D.I. Obstacles to efficient daily meetings in agile development projects: A case study. In ACM/IEEE International Symposium on Empirical Software Engineering and Measurement, October 2013. [54]
PS3	Lalsing, V.; Kishnah, S.; Pudaruth, S. People factors in agile software development and project management. International Journal of Software Engineering & Applications, 2012,3(1), p. 117. [55]
PS4	Stray, V.; Sjøberg, D.I.; Dybå, T. The daily stand-up meeting: A grounded theory study. Journal of Systems and Software, 2016, 114, pp. 101–124. [56]
PS5	Stray, V.; Moe, N.B.; Bergersen, G.R. Are daily stand-up meetings valuable? A survey of developers in software teams. In International Conference on Agile Software Development. Springer, Cham, April, 2017. [57]
PS6	Miller, G.J. Agile problems, challenges, & failures. Paper presented at PMI® Global Congress 2013—North America, New Orleans, LA. Newtown Square, PA: Project Management Institute, 29 October 2013. [58]
PS7	Stray, V.; Moe, N.B.; Sjøberg, D.I. Daily stand-up meetings: start breaking the rules. IEEE Software, 2018, 37(3), pp. 70–77. [59]
PS8	Cruzes, D.S.; Moe, N.B.; Dybå, T. Communication between developers and testers in distributed continuous agile testing. In 2016 IEEE 11th International Conference on Global Software Engineering (ICGSE), August, 2016. [60]
PS9	Amorim, L.F.; Marinho, M.; Sampaio, S. How (un) happiness impacts on software engineers in Agile teams? International Journal of Software Engineering & Applications (IJSEA), 2020, 11(3). [61]
PS10	Unger-Windeler, C.; Klünder, J.A.-C.; Reuscher, T.; Schneider, K. Are Product Owners communicators? A multi-method research approach to provide a more comprehensive picture of Product Owners in practice. Journal of Software: Evolution and Process, 2020, 33(1). [62]
PS11	McHugh, O.; Conboy, K.; Lang, M. Agile practices: The impact on trust in software project teams. IEEE Software, 2011, 29(3), pp. 71–76. [63]
PS12	Westman, M. Scrum—From hype to disappointment? Thesis for the Degree of Bachelor of Science in Computer Science 15.0 credits. Mälardalen University School of Innovation Design and Engineering Västerås, Sweden. 24 May 2023. [64]

Table 2. *Cont.*

Primary Studies (PS)	References
PS13	Stadler, M.; Vallon, R.; Pazderka, M.; Grechenig, T. Agile distributed software development in nine central European teams: Challenges, benefits, and recommendations. <i>International Journal of Computer Science & Information Technology (IJCSIT)</i> , 2019, 11(1), pp. 1–18. [65]
PS14	Mortada, M.; Ayas, H.M.; Hebig, R. Why do software teams deviate from scrum? reasons and implications. In <i>Proceedings of the International Conference on Software and System Processes</i> , Republic of Korea, Seoul, 26–28 June 2020. [40]
PS15	Guillot, I.; Paulmani, G.; Kumar, V.; Fraser, S.N. Case studies of industry-academia research collaborations for software development with agile. In <i>CYTED-RITOS International Workshop on Groupware</i> . 9 July 2017. [66]
PS16	Hajjdiab, H.; Taleb, A.S. Adopting agile software development: issues and challenges. <i>International Journal of Managing Value and Supply Chains (IJMVSC)</i> , 2011, 2(3), pp. 1–10. [67]

Table 3 presents the identified problems along with the primary studies that were examined to elucidate them.

Table 3. Problems in primary studies.

Problems	Primary Studies (PS)
Meetings take too long	PS1, PS2, PS3
Meetings are held frequently	PS2, PS4, PS5, PS6
Some team members are given little attention during meetings	PS2, PS3, PS7, PS8
Communication issues	PS1, PS2, PS7, PS3, PS9, PS10
Negative attitude towards meetings	PS2, PS4, PS7, PS11, PS12
Losing focus	PS2
Some team members do not actively participate in the meeting	PS2
Less time spent on regular tasks	PS2, PS7, PS11
Trust between team members	PS7, PS3, PS13, PS11
Scheduling of tasks	PS2
Motivation for work	PS14, PS15
Avoiding meetings	PS3, PS16, PS12

After analyzing the literature, a theoretical model was developed that represents team problems in agile meetings. Meeting problems and meeting-related team problems are represented in the model in Figure 2. Relationships between meeting issues and team issues are highlighted with blue arrowed lines. Meeting questions are marked with light orange ellipses, and team questions are marked with dark blue ellipses. Problems from one group are also interconnected. Problems in meetings are connected by dark blue lines with arrows in both directions, while problems of a team nature are interconnected by a green line with an arrow in the agreed direction. The authors proposed additional links within the theoretical framework to better capture the complexity of the researched problem. Dashed burgundy arrows indicate relationships between teams' problems. In team dynamics, motivation and trust share a deep connection, where trust acts as a cornerstone, fostering collaboration, communication, and unity among team members and leading to heightened motivation and performance. For teams aspiring to optimize their effectiveness and attain success, nurturing trust emerges as a key priority. Organizations can cultivate a motivating environment among their workforce by aligning task schedules with individual aspirations, offering clarity and autonomy, and acknowledging accomplishments [68]. Moreover, negative perceptions towards meetings can hinder active participation, stemming from various factors like perceived relevance, dynamics of dominance, fear of judgment, and

organizational culture. Tackling these negative attitudes and fostering a positive meeting environment is pivotal for enhancing participation and boosting meeting effectiveness.

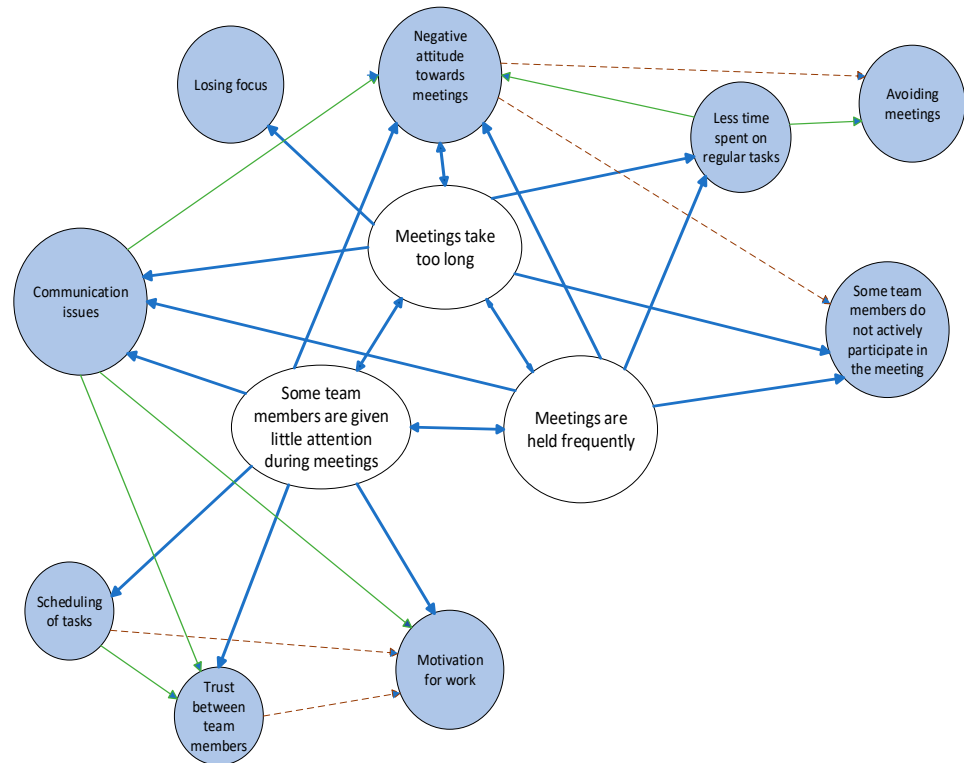


Figure 2. A theoretical model based on agile meeting problems and team issues.

Applying an SLR, it was noticed that meetings are held longer than planned according to the agenda, and the frequency of the meetings is noticeable to a large extent, as well as the fact that some team members do not receive adequate attention during the sessions. These three items form the basis of the model. SLR enabled a detailed analysis of the existing research and the observation of frequent problems, such as the duration and frequency of the meeting and insufficient attention paid to the team members at the meeting. Several studies highlight the prevalence of long meetings that employees attend only to make a good impression on the meeting leader [45]. Employees being late for the meeting automatically indicates an extension of the meeting [53]. In Lalsing's study [55], respondents indicated that meetings often went over schedule due to challenges in reaching agreements on various issues. There is a problem in several studies [56–58] where meetings are held too often, causing dissatisfaction among participants. Allocation of attention within meetings also emerged as a significant item, with certain team members receiving disproportionate focus [54]. In the findings of Lalsing et al. [55], only a select few employees dominated the discussions, while the rest were just listeners. More recent research by Stray et al. [59] indicates that managers predominantly hold status update meetings, where they neglect problem-solving discussions that are important for team motivation. Similarly, Cruzes et al. [60] highlight the problem of getting feedback during meetings due to the unavailability of developers. Different working hours among team members hinders effective communication, exacerbated by physical distance and competing work responsibilities. In some cases, testers have limited participation in meetings and are excluded from the decision-making process. If every member does not have the opportunity to contribute to the meeting, it can affect the success of the project.

The process began with identifying central problems related to agile meetings through a systematic literature review. After establishing these central issues, related problems were identified to understand their impact on team dynamics. It has been observed that long meetings often cause team members to lose focus. It was noticed that team members spend

less time on regular meetings and that some team members do not actively participate in the meetings because they have a negative attitude towards the meeting, which is why they skip meetings. The length of the meeting, therefore, also affects communication. Also, it was observed that meeting participation can reduce the time required for the preparation of regular tasks and that this leads to a negative attitude towards the meeting. Some team members then do not actively participate in the meeting. Therefore, it affects communication between them. Some team members receive less attention during the meeting, which automatically results in poorer communication in the meeting, and a negative attitude towards the meeting can result from this. This problem can affect the motivation to work and the trust between team members, as well as the assignment of tasks between team members. If communication is not good, it can affect motivation and trust between them. From all related problems, key issues were selected for potential solutions based on their criticality to team productivity and the feasibility of impact through a technical solution. The model focuses on addressing specific challenges rather than all problems at once. In particular, the emphasis is on problems like meeting duration, meeting frequency, and insufficient attention during meetings, as these directly influence the quality and efficiency of agile meetings. Addressing these issues can significantly enhance overall productivity and positively impact the company's bottom line. The model serves not only to identify problems but also to be a good basis for solving them. Below this text is Figure 3, which schematically presents the step-by-step process of how the model was created.

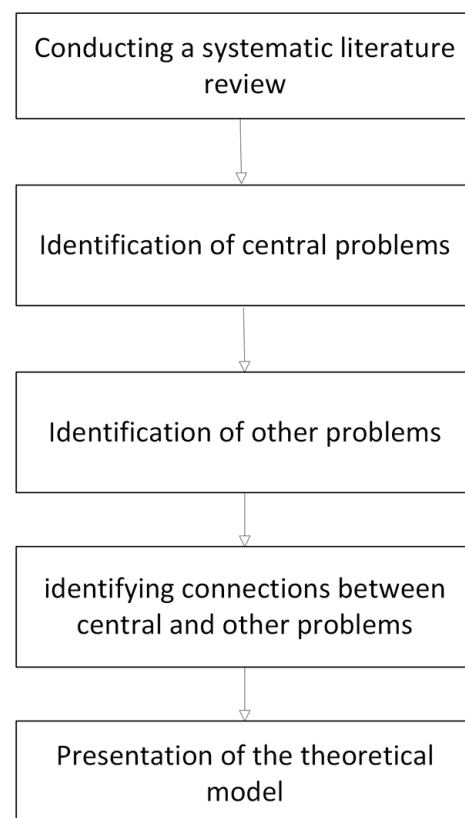


Figure 3. Presentation of the model creation process.

4. Validation of Theoretical Model in a Qualitative Study

Validation of the model developed through a systematic literature review (SLR) is crucial. This validation process involves gathering feedback from IT experts on aspects they consider important. The participants were shown the model and invited to give suggestions for its improvement. Therefore, this validation aims to assess whether the corresponding theoretical model needs to be improved and what modifications are needed.

4.1. Sampling of Participants

In a qualitative study involving the validation of a theoretical model derived from SLR, the sampling of participants plays a crucial role in ensuring the robustness and credibility of the validation process [69]. In qualitative research, an interview is a data collection method that involves talking with participants to gather rich, detailed information about their experiences, perspectives, beliefs, and behaviors. Interviewing is one of the most common methods used in small-scale research [70]. We purposefully selected the participants for the validated study based on their expertise and relevance to the subject of the model, as in the authors' study [23]. Seven practitioners from the IT industry with work experience ranging from three to eleven years were included in the research to help improve the presented model. All the interview participants were programmers at middle and senior levels in their respective companies, as shown in Table 4. These participants were chosen for their extensive work experience. Their expertise allows them to provide essential feedback on the presented model. Their suggestions are invaluable in refining and enhancing the model, ensuring it becomes more complex and of higher quality. In addition to selecting individuals with significant experience, these participants are also trustworthy and able to express their own opinions. The sample size depends on the population being studied and its size and composition, and since it is a case study, we have enough respondents to make suggestions for the presented model. Qualitative research usually involves relatively few people and tends to be associated with smaller-scale studies [71]. Marshall et al. [72] and Hennink et al. [73] state theoretical arguments for sample size estimation. The concept of data saturation, which applies to all qualitative research that uses interviews as the primary source of data, implies the constant questioning of new participants until enough data is collected, such that it begins to repeat itself. Saturation is achieved when the researcher collects data when nothing new is added [74,75]. The assessment of adequate sample size is directly related to the concept of saturation [76]. The saturation of constructs is used by qualitative research theorists to verify the rigor and validity of a study [77]. "Small sample sizes, such as six to ten participants, are common and sufficient for providing rich data for thematic analysis" [47]. *In addition, methodological literature also suggests that in qualitative research, an acceptable number of participants is between 5 and 25, depending on the complexity of the research question and the study design* [48]. In this study, the authors believe that enough practitioners were interviewed because the existing problems and connections are repeated. It was agreed that participants would be anonymous to encourage an open and honest discussion, which, in turn, would improve the quality of the data collected [78]. We followed ethical principles by ensuring informed consent, confidentiality, and respect for participants' autonomy. Participants were given clear information about the study's aims, procedures, and the potential implications of their participation [21].

Table 4. Work experience and seniority/level of participants.

Participants	Work Experience	Seniority/Level in Company
1	9 years	Senior
2	8 years	Senior
3	4 years	Medior
4	3 years	Medior
5	6 years	Senior
6	5 years	Medior
7	11 years	Senior

4.2. Data Collection

Data collection in this study involved conducting individual, in-depth, semi-structured interviews, which were recorded and transcribed. We opted for interviews for two primary reasons. Firstly, the semi-structured format allows for addressing ambiguities in questions

directly. Secondly, it allows the interviewers to observe the participants and encourage them to answer all the questions well and thus provide better insight [79].

Transcription, a process that involves careful listening and/or viewing, is the initial step in data analysis [80]. Semi-structured interviews are useful when most questions require follow-up questions. We wanted to allow participants to express their opinions [81]. The interviews were conducted in person and via the Zoom platform, depending on the circumstances but also on the wishes of the participants. Interviews via an online platform offer advantages such as convenience, efficiency, and flexibility [82]. Before participation, participants were fully informed about the aims of the study and received informed consent forms to ensure voluntary participation and the confidentiality of their responses.

4.3. Data Analysis

Qualitative methods offer a comprehensive approach to exploring intricate issues, such as those highlighted within our model and the valuable insights gleaned from interviews with IT experts. It is noteworthy that in qualitative studies, data analysis often commences during the data collection phase [83]. We conducted a qualitative analysis to delve into participants' perspectives, potentially refining the model based on their feedback. Transcriptions of participants' views were diligently documented in a Word document [84], with opinions suggesting potential modifications thoughtfully integrated into the evolving model. In qualitative research, various data analysis strategies are typically employed, with researchers often combining several approaches within a single study [84–86]. Following the methodological footsteps of Gupta et al. [87], our paper presents a concise literature review to systematically elucidate the challenges faced by agile meeting teams. We synthesized these challenges into a model, subsequently validating it through interviews with seasoned developers, aligning with Gupta et al.'s approach [87]. Echoing Senapathi's methodology [88], upon the completion of interviews and the presentation of participant perspectives, our model underwent refinement. This process, akin to Nurdian's study [89], involved comparing literature-derived questions on agile meetings with insights gleaned from participant interviews.

Thematic analysis played a crucial role in our study, revealing novel issues and connections previously unaccounted for in the existing model. This systematic approach, as outlined by Braun and Clarke [90], enabled us to meticulously examine interview data, incorporating participants' suggestions and insights to enrich the model's scope and applicability.

Our analysis of qualitative data was rigorous, primarily following Reyes' coding method [91], which is recognized for its efficacy in uncovering underlying messages and patterns, as highlighted by Saldana [92]. We then synthesized the coded data into themes. Thematic analysis, advocated by Guest et al. [93], proved invaluable in identifying patterns and insights, enhancing our theoretical model's depth and clarity. In summary, our analysis involved coding the data, deriving participant opinions, and synthesizing codes into themes, employing a structured six-stage methodology [94]. The themes were identified as follows:

1. **Challenges with Daily Standup Meetings:** This theme focuses on the difficulties arising from the daily standup meetings, such as team members not having new updates due to longer task durations and the resulting lack of motivation and negative attitudes toward the meetings.
2. **Impact of Meeting Frequency on Focus and Motivation:** This discusses how frequent meetings can lead to a loss of focus, decreased motivation, and even skipping meetings altogether.
3. **Consequences of Lost Focus and Motivation on Task Completion:** This addresses how decreased concentration in meetings can result in less time being spent on tasks and reduced productivity.
4. **Relationship Between Motivation and Active Participation:** This discusses how low motivation can lead to passive participation or even complete disengagement from meetings.
5. **Communication Breakdown Due to Non-Participation:** This discusses how inactive participation can lead to communication breakdowns and hinder collaboration.

These themes highlight the interconnected nature of meeting dynamics, motivation levels, task completion, and communication effectiveness within the team context. The theoretical model provided a structured basis for interpreting these themes, with modifications made following a thorough qualitative analysis. In the revised model, additional connections were incorporated, along with a new problem item marked with specific colors to denote each participant's contribution. Following a thorough qualitative analysis, refinements were made to the model depicted in Figure 4. In the modified model, additional connections are shown, and the problem is marked with a certain color. Each participant who added specific connections/issues between items is marked with a 'P' and a serial number. A connection, added by participant number two, is shown in gray. The connection added by participant number three is shown in orange. The connection of the fourth participant is shown in purple, while the connection of the fifth participant is highlighted in red. The issue titled "Some team members don't have new updates during the meeting", added by the second participant, is highlighted with gray ellipses. The first participant, the sixth participant, and the seventh participant agreed with the preliminary model and had nothing new to add to the existing model. At the end of the interview, the seventh participant summarized all the items from the model once more to make sure that he had not forgotten to add anything. The seventh participant emphasized that meetings should be as short, infrequent, and purposeful as possible. They should provide useful information, solve problems, or offer help rather than being pointless discussions that only tire out team members. Good team relationships are crucial, and every member should be heard and given opportunities to grow, regardless of how vocal they are. It is also important to value all roles equally, including testers. In summary, meaningful and efficient meetings, fair task distribution, and good team dynamics are key to successful agile practices.

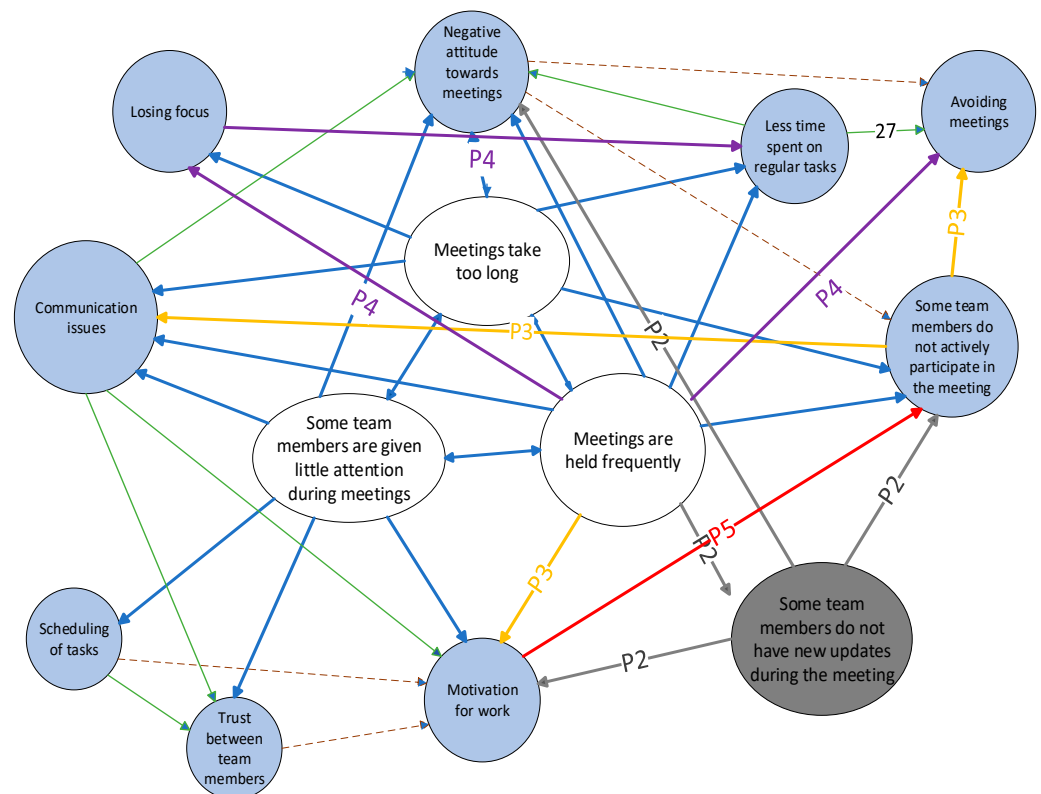


Figure 4. Improved theoretical model.

5. Discussion of an Improved Theoretical Model

In the interview that was conducted as part of this research, we had seven participants, in which the participants gave suggestions for modifying the model if they thought that modification was needed. They suggested what should be added to the problems, what

connections should still exist between the problems, etc. They presented their experiences with these problems and how they look at these problems from their point of view based on many years of experience in the company. It can be said that most of them agreed with the presented model, and a few of them suggested other significant problems to add to the model. For the most part, the participants considered that all the mentioned problems were important to be included in the model. P#2 stated that he does not like being called to meetings several times a day. It happens that some team members have nothing new to add in the meeting because they are still working on the same task, and the work on the task often lasts for several days; this leads to the engineer having nothing new to say, which develops negativity towards the meeting.

“It is not good when the management call to some meetings several times during the day, so several hours a day are spent in different meetings, which leads to the exhaustion of team members”. It points to potential changes in the model: “Meetings are held frequently” should perhaps be linked to the new issue “some team members do not have new updates during the meeting”. Maybe it’s already included through some problems, “Meetings are held frequently” should be linked to that new problem, and that problem should be linked to the problem of “team members not being active in meetings”. As a result, team members waste time on regular tasks. Perhaps even the new problem should be linked to “motivation” or even better to “negative attitude towards the meeting”.

P#3 stated that a connection needs to be made between meeting skipping and people not actively participating. He also stated that motivation is connected with the fact that meetings are held frequently. *“When a week is arranged for me to have many meetings, even if they are important meetings, if there are many, it simply cannot help but have a bad effect on motivation. I think that communication should be connected with the fact that some team members do not actively participate in meetings. If someone does not actively participate in meetings, that is specifically a problem in communication and leads to even bigger problems in communication. Communication is one of the most important factors because, for example, we linked it to motivation and to negative attitudes towards meetings and to trust. Communication is the main thing that must be good in a team for a team to work well. I think some routine meetings should be reduced to a written form in the form of a written survey where people can get questions to answer and it all reaches the manager at a certain time where the manager will see everything he wants and I will be able to read everything. And then he won’t spend so much time in meetings, while everyone has the opportunity to write what he has to say. Depending on the work dynamics the relationship of the team members and the communication in the team, it depends on what types of meetings should be held in writing and person. Each team could write trivia meetings for themselves and it depends on the team”.*

The next participant, P#4, stated that “loss of focus” should be associated with “less time spent on regular tasks” because if one does not have focus, one has no will to work, regardless of meetings. *“Maybe connect that ‘meetings too often’ with ‘loss of focus’ because if you have meetings every two hours and you have to do something for a long time, you need better focus. If several times a day several meetings don’t go in order, and you have breaks between meetings, say an hour, you cannot perform your regular tasks attentively. If meetings are held frequently, meetings are skipped”.*

Participant P#5, who made some suggestions, stated that he would associate motivation with people who do not participate in meetings because he believes that if an employee is not motivated to work, he may not even be interested in contributing to meetings. He also stated that sometimes there are situations where some problems are discussed several times in a meeting and the problems are not solved, and this can lead to people not being motivated and participating less in meetings because they feel that the meetings are losing their meaning. As an example, he stated that *“people complain that meetings take a long time, and despite this being recognized as a problem, meetings still take a long time, so once you mention the problem once or twice, you don’t want to repeat it the next time”.*

To summarize and present the feedback from the interview participants regarding the model of agile meetings, the following Table 5 can be used. It highlights the key suggestions and observations made by the participants. Only the respondents’ suggestions and observations are shown, i.e., if the participants said that something could be improved in the model.

Table 5. Presentation of feedback from interview participants who gave suggestions and observations for the presented model.

Participant	Key Suggestion	Comments and Observations
P#1	Supports the model without the proposed modifications	Fully agrees with the presented model and has no additional suggestions for changes.
P#2	Link “Meetings are held frequently” with “some team members do not have new updates during the meeting”	Frequent meetings can lead to team members having nothing new to report, which contributes to negativity and exhaustion. Suggested connecting this with “Some team members do not actively participate in the meeting” and “Motivation for work”.
P#3	Connect communication issues with meeting frequency and meeting participation	Frequent meetings negatively impact motivation for work. Suggested using written forms for routine meetings to improve efficiency.
P#4	Associate “loss of focus” with “less time spent on regular tasks”	Frequent meetings disrupt focus and lead to reduced productivity. Suggested linking “Meetings are held frequently” with “Losing focus” to address the impact on task performance.
P#5	Link motivation with participation	Lack of motivation affects participation. Repeatedly discussing unresolved problems can demotivate team members. Suggested improving meeting effectiveness to maintain motivation.
P#6	No additional suggestions for modifications	Did not provide any new suggestions or feedback. Maintains alignment with the existing model.
P#7	No additional suggestions for modifications	Did not provide any new suggestions or feedback. Maintains alignment with the existing model.

6. Implications

The authors offered a concise, systematic review of the literature that was the basis for the creation of the model. The model was presented to the participants and refined through discussions. The improved model highlights issues that arise in meetings. If problems go undetected, problems can foster dissatisfaction among team members, ultimately affecting project success and overall company performance. The authors can create a technical solution aligned with a custom model to facilitate specific types of meetings.

This paper offers scientific researchers an insight into the methodology and presentation of the results of the review. In particular, the results are presented through a model depicting the challenges of meetings. Researchers can follow the progress of the model as it is refined through interviews with respondents, but they can also use the improved model as a basis for development and, through subsequent discussions with IT experts, create improved versions of the model. Researchers can see the importance of interviews in qualitative research and realize that interviews can provide a richer insight into certain phenomena they are studying since there is great flexibility in the research; respondents can ask additional questions during the conversation, and thus, more detailed answers can be obtained. Researchers can use the methodological approaches used in the study and determine strengths, limitations, and areas for improvement. Practitioners can create technical solutions to simplify the meeting process. In this way, they would contribute to the IT industry by improving team member collaboration and overall productivity. The findings of the study may open opportunities for collaboration between researchers, practitioners, and other stakeholders.

7. Trustworthiness

Trustworthiness encompasses the credibility, reliability, and dependability of research findings and interpretations, instilling confidence in the accuracy, integrity, and transparency of the research process and outcomes. The subjective nature of data collection and interpretation demands meticulous attention to ensure the research's credibility and trustworthiness [90].

A small sample size may be sufficient when methodological issues and objectives are precisely defined [71]. To establish credibility, we employed various strategies to accurately represent the perspectives of IT experts. Conducting semi-structured interviews allowed the participants to express their thoughts and experiences freely. We had a good relationship with the participants before and during the interview. The interviewees spoke openly with us. The authors of this paper actively engaged in discussions and the development of findings, ultimately reaching agreement on defined themes. This underscores the collaborative nature of the analysis process, enhancing the confidentiality of the study findings.

We believe that the insights gained from our study are useful to other IT companies. We followed a systematic approach to data collection and analysis to ensure the reliability of the study. Also, we followed the guidelines for conducting the thematic analysis and described the complete process. We tried to transcribe the interviews verbatim. All misunderstandings were resolved through discussions between the authors.

8. Conclusions

This study aimed to develop a theoretical model related to issues in agile meetings and team dynamics. The theoretical model was developed through a systematic literature review and enriched based on qualitative analysis. The systematic review provided an understanding of the existing research on agile meetings, which was key to developing the initial version of our model. This preliminary model, based on the latest knowledge, served as the basis for further research. We presented the initial model to IT experts to validate its coverage. Feedback from these experts highlights strengths and areas for improvement. The authors highlighted the following additions to the model: It is necessary to account for the issue where team members have nothing new to say in daily standup meetings because they work on tasks for several days. This can lead to a lack of motivation to work and a negative attitude towards the meetings. The frequency of the meetings can cause a loss of focus and motivation, resulting in skipped meetings. If focus is lost and concentration drops during the meetings, it can lead to employees spending less time on their regular tasks. A loss of motivation can result in employees not actively participating in meetings. If employees have nothing new to contribute, they may become inactive, which affects communication and can lead to skipping meetings altogether.

Therefore, we have developed a model that is in line with the scientific literature but also with the needs of participants who are part of agile teams in IT companies.

In the future, the authors will create a survey based on the modified model to identify as many problems as possible in agile meetings. A quantitative analysis of the survey data is planned to increase confidence in the results of the analysis through different analyses of the same model. Also, future work will focus on the development of software tools by IT experts that will facilitate the implementation of meetings. The developed software solution will try to solve the most important problems identified in the model, which would represent a key contribution to practice. The plan for the future is to develop and validate more software solutions in industrial practice.

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